

Using Machine Learning to Classify EEG Data Collected with or without Haptic Feedback During a Simulated Drilling Task

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ABSTRACT

Simulation environments (SE) are becoming important tools that can be leveraged to implement training protocols and educational resources. Electroencephalography (EEG) is used to compare the effects of different types of feedback in SE, but it can be challenging to know which aspects of EEG represent the impact of different types of feedback on neural processing. For this study, the Minimum-Redundancy-Maximum-Relevance (MRMR) algorithm, and the MRMR in conjunction with the Mann-Whitney U statistical test were applied to select the most relevant EEG features associated with haptic and non-haptic feedback in a simulated drilling task. EEG was analyzed based on the extraction and selection of different machine learning features.

Index Terms: Human-centered computing—Human computer interaction (HCI) —Interaction devices—Haptic devices.

1 INTRODUCTION

Artificial Intelligence (AI), computer simulations, and virtual reality (VR) are increasingly being leveraged to create and implement training protocols and educational resources. VR and simulations may be particularly important for high-risk occupations that have limited opportunities for real-world training, and they also have application to rehabilitation [1]. Developing an enhanced understanding of physiological constraints associated with VR is fundamental to inform the way in which VR training is developed and implemented. Biological signals, such as those obtained using electroencephalography (EEG), show promise as an objective means of assessing sensory and neural processing associated with task performance in virtual environments [2]. However, the current research that has assessed EEG recorded in these settings has resulted in heterogeneous findings. Specifically, there is limited research that has elucidated specific EEG markers that can reliably discriminate the impact of haptic feedback in virtual environments.

This paper focuses on differentiating EEG signals representative of neural circuitry associated with haptic and non-haptic sensory feedback when completing a simulated drilling task and assess the ability of machine learning approaches to identify features of EEG data related to haptic feedback.

2 METHODS

The EEG recordings used were collected at the Laboratory of Human Neurophysiology and Rehabilitation, Ontario Tech University, Oshawa, Canada (REB approval 15402). The participants (n=15) were healthy young adults, 22.9 ($\pm$ 1.4) years of age, inclusion criteria required right handedness (confirmed by the Edinburgh Handedness Questionnaire) and prior experience using a drill. Prior to the start of the experiment, participants completed a familiarization phase, performing multiple trials of both haptic and non-haptic conditions. After completing the familiarization phase, each participant completed a total of 200

pseudo-randomized trials: 100 haptic and 100 non-haptic. In both conditions, auditory stimuli were consistent, while force-feedback was only provided during the haptic condition [3]. Once the dataset had been obtained all blocks of EEG data were grouped according to their class (haptic or non-haptic). As indicated by previous research, nine electrodes are considered particularly relevant to the analysis of event-related desynchronization in sensorimotor paradigms [3]; F3, Fz, F4, C3, Cz, C4, P3, Pz, and P4. We also analyzed four different frequency bands: Low Alpha (8-9.5 Hz), High Alpha (10-13 Hz), Low Beta (13-21 Hz), and High Beta (21-30 Hz), as the activation of these bands is related to the response to various tasks, and is thought to signify attentional processes and general task demands [3]. Different types of features were extracted based on previous work focused on EEG features extraction [4]. Hjorth parameters (time domain), Power Spectral Density (PSD), Spectral entropy, Permutation entropy, Higuchi's Fractal Dimension, Hurst's Exponent, Skewness and Kurtosis.

Two different groups of relevant features were created in relation to the technique applied, the first technique was the Minimum-Redundancy-Maximum-Relevance (MRMR) method. This method uses the relationship between the F-statistic calculated for each feature with the target variable, and the Pearson correlation applied to each individual feature with the rest of the features [5]. The second technique was MRMR combined with the Mann-Whitney U statistical test, which was used to select the relevant groups where the features that showed statistical differences [6]. The classification stage was performed using the feature sets generated by each feature selection proposal. Seven classification models were implemented: Stochastic Gradient Descent (SGD), SVM, Decision Tree, Naive Bayes, KNN, random forest and MLP. A heuristic method was applied to find the training hyperparameters of the models. 80% of the data set was used for training and the remaining 20% was used to test the models. The results obtained allowed the specific analysis of the 10 most relevant features, and the EEG channels and frequency bands that provided the most relevant information to differentiate the signals where haptic feedback was present.

3 RESULTS

Ten features were extracted from nine EEG channels and four different frequency bands. In total 360 features were obtained. In order to determine the relative weighting of each feature to the model, the 20 most relevant features were determined using MRMR. After applying the Mann-Whitney U test to the first dataset 14 features showed statistical differences ($p < 0.05$). Classification of both data sets was performed using different batches. The first classification was used as an input for training the models using the 5, 10, 15, and 20 most relevant features according to the relevance score yielded by MRMR and the total set of extracted features. For

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the second dataset 5, 10, and 14 features were used to train the models. Topographic maps were generated in order to visualize the behavior of all channels with respect to the numerical value of the features and frequency bands that provide the most information. The intensity of the color scale shows the average value of a channel for a specific feature in a frequency band as indicated by the scales included with each map (Figure 1). This approach accurately differentiated haptic feedback processing from non-haptic feedback processing using our combination for feature selection with the 10 most relevant features and achieving up to 99% accuracy (Table 1). The electrodes with the most relevant information were Cz, Pz, Fz, P4 and C3. The Hurst exponent in the 13 - 21 Hz frequency band had the midline and P4 as relevant channels. The kurtosis had two relevant frequency bands low beta and high beta, with Cz and F4 as the relevant channels. The Power Spectral Density (PSD) and variance (activity according to Hjorth parameters) had the same frequency bands and relevant channels, Fz for high alpha and Cz for low beta. Finally, Spectral Entropy for low beta had C3, Cz, and Pz as relevant channels. All channels showed higher mean values for non-haptic feedback processing, except the C3 channel for Spectral Entropy at low beta.

Classifier	5 Best Features	10 Best Features	14 Best Features
Stochastic Gradient Descent (SGD)	66	71	71
Support Vector Classifier (SVC)	84	98	99
Decision Tree	88	98	96
Naive Bayes	64	65	67
K - Nearest Neighbors (KNN)	90	99	99
Random Forest	90	99	100
Multi-Layer Perceptron (MLP)	91	99	99

Table 1: Percentage of accuracy for each classifier using the second data set as inputs.

4 CONCLUSION

This study is the first to our knowledge to utilize machine learning approaches to sort neurophysiological data extracted from EEG signals when processing haptic sensory feedback. The MRMR approach proved useful, as with only 20 features accuracy percentages were almost equal to those obtained using all 360 features. Similarly, the use of the Mann-Whitney U test was complementary to MRMR, allowing a reduction to 10 features that could be used to generate models with better performance. The second feature selection method is a novel finding, and we are not aware of any previous work that has used the same technique to classify EEG signals collected with and without haptic feedback. The generated models yielded promising results, since in addition to surpassing the results of previous work, it also demonstrates that they can be effective tools to identify EEG signals associated with haptic feedback. In terms of the behavior the higher mean values for non-haptic feedback processing, (except C3 channel Spectral Entropy at low beta) suggests that non-haptic feedback processing demonstrates higher variance, stability, and power in the frequency bands associated with mental activity. This may be because drilling is a task normally associated with haptic feedback and in its absence, the individual requires greater neural processing to complete the task. These results are promising and have potential to inform future applications transforming the way EEG data analysis is approached and results interpreted, while using minimal features. Future work should use a bigger data set that allows to

implement a validation system confirmed by training, validation, and test groups.

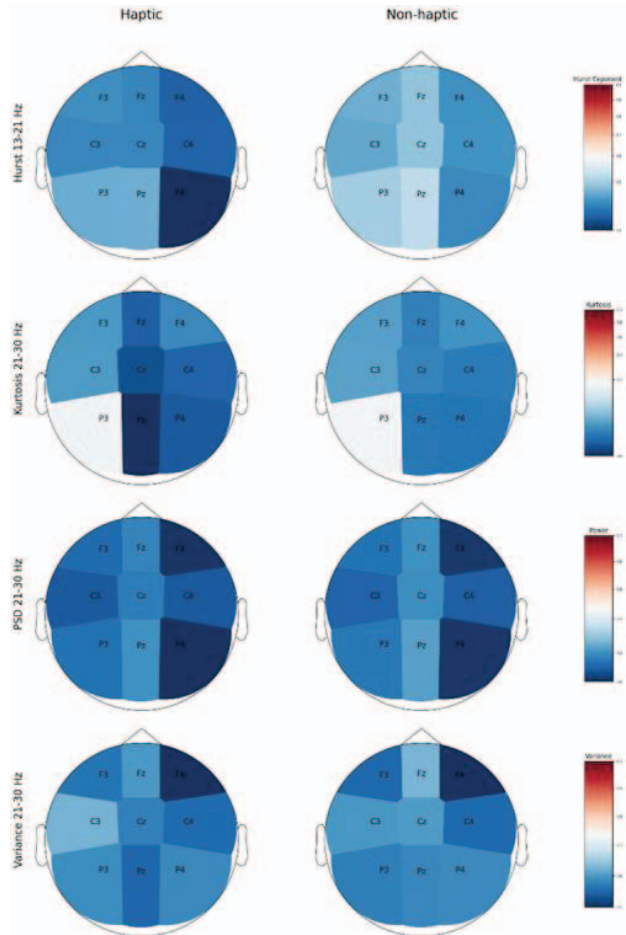


Figure 1: Topographic map of the most relevant features and frequency bands.

REFERENCES

- [1] V. Matijević, A. Šečić, V. Mašić Fabac, M. Sunić, Z. Kolak, y M. Znika, «Virtual reality in rehabilitation and therapy», *Acta clinica Croatica*, vol. 52, pp. 453-7, dic. 2013.
- [2] M. Athif *et al.*, «Using Biosignals for Objective Measurement of Presence in Virtual Reality Environments», en *2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, Montreal, QC, Canada: IEEE, jul. 2020, pp. 3035-3039. doi: 10.1109/EMBC44109.2020.9176022.
- [3] B. Grant, «Behavioral and neurophysiological measures of haptic feedback during a drilling simulation», Doctoral dissertation, Ontario Tech University, Canada, 2019.
- [4] I. Stancin, M. Cifrek, y A. Jovic, «A Review of EEG Signal Features and Their Application in Driver Drowsiness Detection Systems», *Sensors*, vol. 21, n.º 11, p. 3786, may 2021, doi: 10.3390/s21113786.
- [5] Z. Zhao, R. Anand, y M. Wang, «Maximum Relevance and Minimum Redundancy Feature Selection Methods for a Marketing Machine Learning Platform», en *2019 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, Washington, DC, USA: IEEE, oct. 2019, pp. 442-452. doi: 10.1109/DSAA.2019.00059.
- [6] H. B. Mann y D. R. Whitney, «On a Test of Whether one of Two Random Variables is Stochastically Larger than the Other», *Ann. Math. Statist.*, vol. 18, n.º 1, pp. 50-60, mar. 1947, doi: 10.1214/aoms/1177730491.